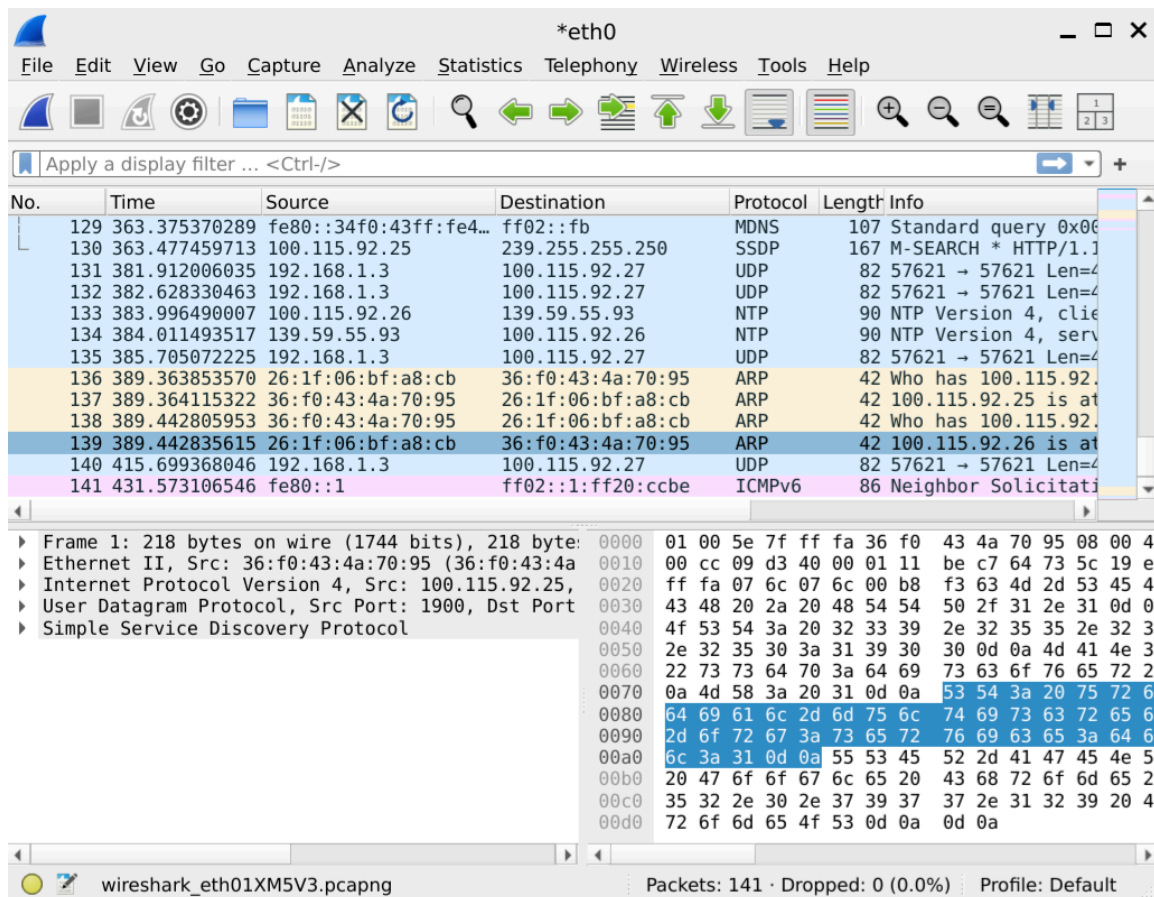


Task 2: Basic Wireshark Traffic Capture & Analysis

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Dept: CSE core B section

1. To capture my own network traffic using Wireshark and identify different protocols.
2. I installed Wireshark and captured traffic using the eth0 interface. I browsed a few websites like wikipedia, google, and searched a few and captured the network packets for a short time. The final capture had 141 packets and 0 dropped packets.



3. Protocols Identified

- **ARP:** Used to find the MAC address related to an IP address on the local network.
- **mDNS:** Used for local name and service discovery. It uses UDP port 5353.
- **NTP:** Used for network time synchronization.

- **SSDP:** Used for discovering devices and services on a local network. It uses UDP port 1900.

Screenshot – ARP packets

task2_capture.pcapng

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arp

No.	Time	Source	Destination	Protocol	Length	Info
11	132.851900377	26:f0:06:bf:a8:cb	36:f0:43:4a:70:95	ARP	42	Who has 100.115.92.25? T
12	132.852247387	36:f0:43:4a:70:95	26:f0:06:bf:a8:cb	ARP	42	100.115.92.25 is at 36:f
13	132.950104870	36:f0:43:4a:70:95	26:f0:06:bf:a8:cb	ARP	42	Who has 100.115.92.26? T
14	132.950136756	26:f0:06:bf:a8:cb	36:f0:43:4a:70:95	ARP	42	100.115.92.26 is at 26:1
136	389.363853570	26:f0:06:bf:a8:cb	36:f0:43:4a:70:95	ARP	42	Who has 100.115.92.25? T
137	389.364115322	36:f0:43:4a:70:95	26:f0:06:bf:a8:cb	ARP	42	100.115.92.25 is at 36:f
138	389.442805953	36:f0:43:4a:70:95	26:f0:06:bf:a8:cb	ARP	42	Who has 100.115.92.26? T
139	389.442835615	26:f0:06:bf:a8:cb	36:f0:43:4a:70:95	ARP	42	100.115.92.26 is at 26:1

Frame 11: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
 Ethernet II, Src: 26:f0:06:bf:a8:cb (26:f0:06:bf:a8:cb), Dst: 36:f0:43:4a:70:95 (36:f0:43:4a:70:95)
 Address Resolution Protocol (request)

0000 36 f0 43 4a 70 95 26 1f 06 bf a8 cb 08 06 00 00
 0010 08 00 06 04 00 01 26 1f 06 bf a8 cb 64 73 5c
 0020 00 00 00 00 00 00 64 73 5c 19

Address Resolution Protocol: Protocol Packets: 141 · Displayed: 8 (5.7%) Profile: Default

Screenshot – mDNS packets

task2_capture.pcapng

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mdns

No.	Time	Source	Destination	Protocol	Length	Info
21	305.412443231	100.115.92.25	224.0.0.251	MDNS	87	Standard query 0x00
22	305.412443361	fe80::34f0:43ff:fe4...	ff02::fb	MDNS	107	Standard query 0x00
24	305.513260599	100.115.92.25	224.0.0.251	MDNS	103	Standard query 0x00
25	305.513590718	fe80::34f0:43ff:fe4...	ff02::fb	MDNS	123	Standard query 0x00
26	305.513590828	100.115.92.25	224.0.0.251	MDNS	103	Standard query 0x00
27	305.513590888	fe80::34f0:43ff:fe4...	ff02::fb	MDNS	123	Standard query 0x00
28	305.513590948	100.115.92.25	224.0.0.251	MDNS	119	Standard query 0x00
29	305.513591019	fe80::34f0:43ff:fe4...	ff02::fb	MDNS	139	Standard query 0x00
30	306.434957575	100.115.92.25	224.0.0.251	MDNS	96	Standard query 0x00
31	306.434957675	fe80::34f0:43ff:fe4...	ff02::fb	MDNS	116	Standard query 0x00
32	306.537682368	100.115.92.25	224.0.0.251	MDNS	119	Standard query 0x00
33	306.537682468	fe80::34f0:43ff:fe4...	ff02::fb	MDNS	139	Standard query 0x00
34	307.358110967	100.115.92.25	224.0.0.251	MDNS	87	Standard query 0x00
35	307.358111057	fe80::34f0:43ff:fe4...	ff02::fb	MDNS	107	Standard query 0x00

Frame 21: 87 bytes on wire (696 bits), 87 bytes captured (696 bits) on interface 0
 Ethernet II, Src: 36:f0:43:4a:70:95 (36:f0:43:4a:70:95), Dst: 01:00:5e:00:00:fb
 Internet Protocol Version 4, Src: 100.115.92.25, Dst: 224.0.0.251
 User Datagram Protocol, Src Port: 5353, Dst Port: 5353
 Multicast Domain Name System (query)

0000 01 00 5e 00 00 fb 36 f0 43 4a 70 95 08 00 45
 0010 00 49 c1 a0 40 00 01 11 16 7c 64 73 5c 19 e0
 0020 00 fb 14 e9 14 e9 00 35 a1 ce 00 00 00 00 00
 0030 00 00 00 00 00 00 10 5f 73 70 6f 74 69 66 79
 0040 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 61
 0050 61 6c 00 00 0c 00 01

Multicast Domain Name System: Protocol Packets: 141 · Displayed: 74 (52.5%) Profile: Default

Screenshot – NTP packets

task2_capture.pcapng

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ntp

No.	Time	Source	Destination	Protocol	Length	Info
9	127.747605596	100.115.92.26	139.59.55.93	NTP	90	NTP Version 4, client
10	127.765571457	139.59.55.93	100.115.92.26	NTP	90	NTP Version 4, server
133	383.996490007	100.115.92.26	139.59.55.93	NTP	90	NTP Version 4, client
134	384.011493517	139.59.55.93	100.115.92.26	NTP	90	NTP Version 4, server

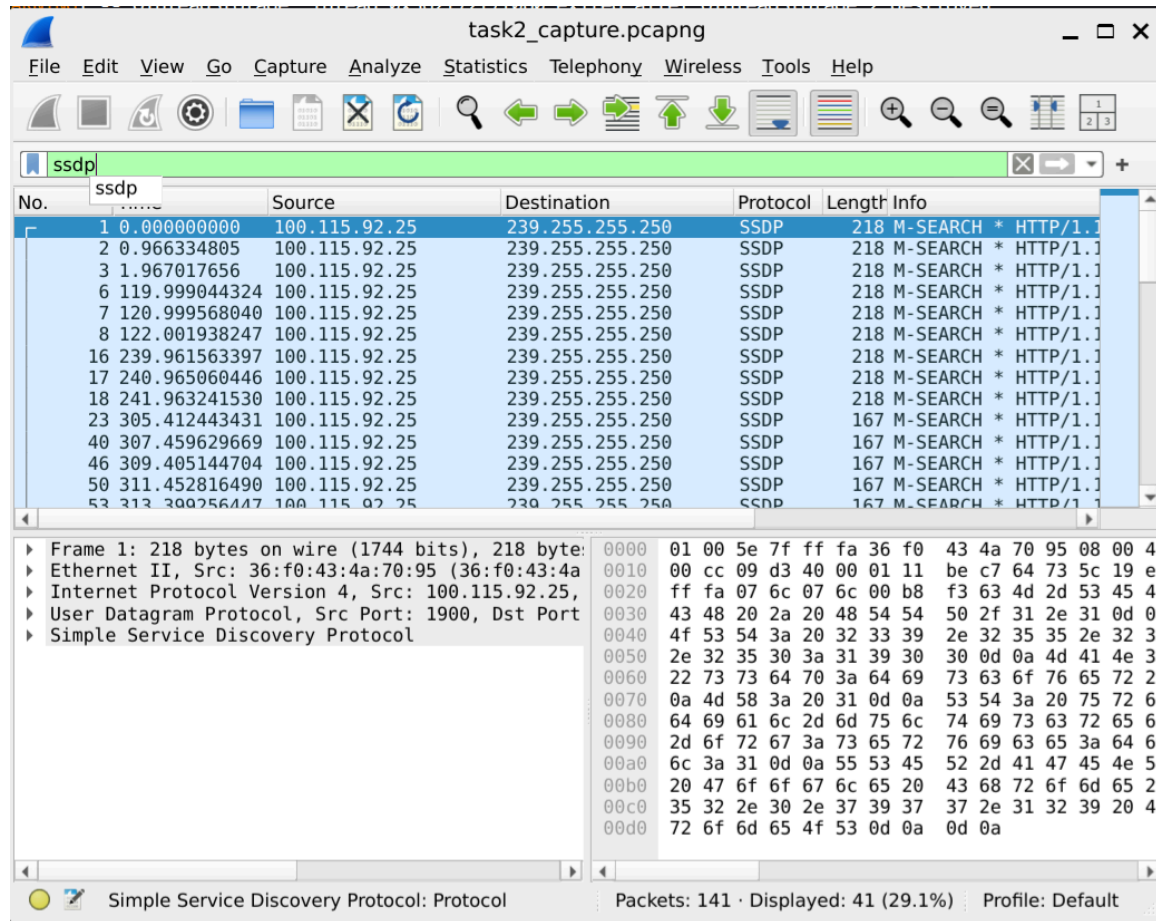
Frame 9: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface 0
 Ethernet II, Src: 26:1f:06:bf:a8:cb (26:1f:06:bf:a8:cb), Dst: 08:00:27:00:00:00
 Internet Protocol Version 4, Src: 100.115.92.26, Dst: 139.59.55.93
 User Datagram Protocol, Src Port: 35797, Dst Port: 123
 Network Time Protocol (NTP Version 4, client)

0000 36 f0 43 4a 70 95 26 1f 06 bf a8 cb 08 00 45
 0010 00 4c d9 28 40 00 40 11 dd 9a 64 73 5c 1a 80
 0020 37 5d 8b d5 00 7b 00 38 83 6f 23 00 00 00 00
 0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 0050 00 00 29 d2 22 04 7c c4 ec 92

Network Time Protocol: Protocol

Packets: 141 · Displayed: 4 (2.8%) Profile: Default

Screenshot – SSDP packets



4. Packet Walkthrough – ARP

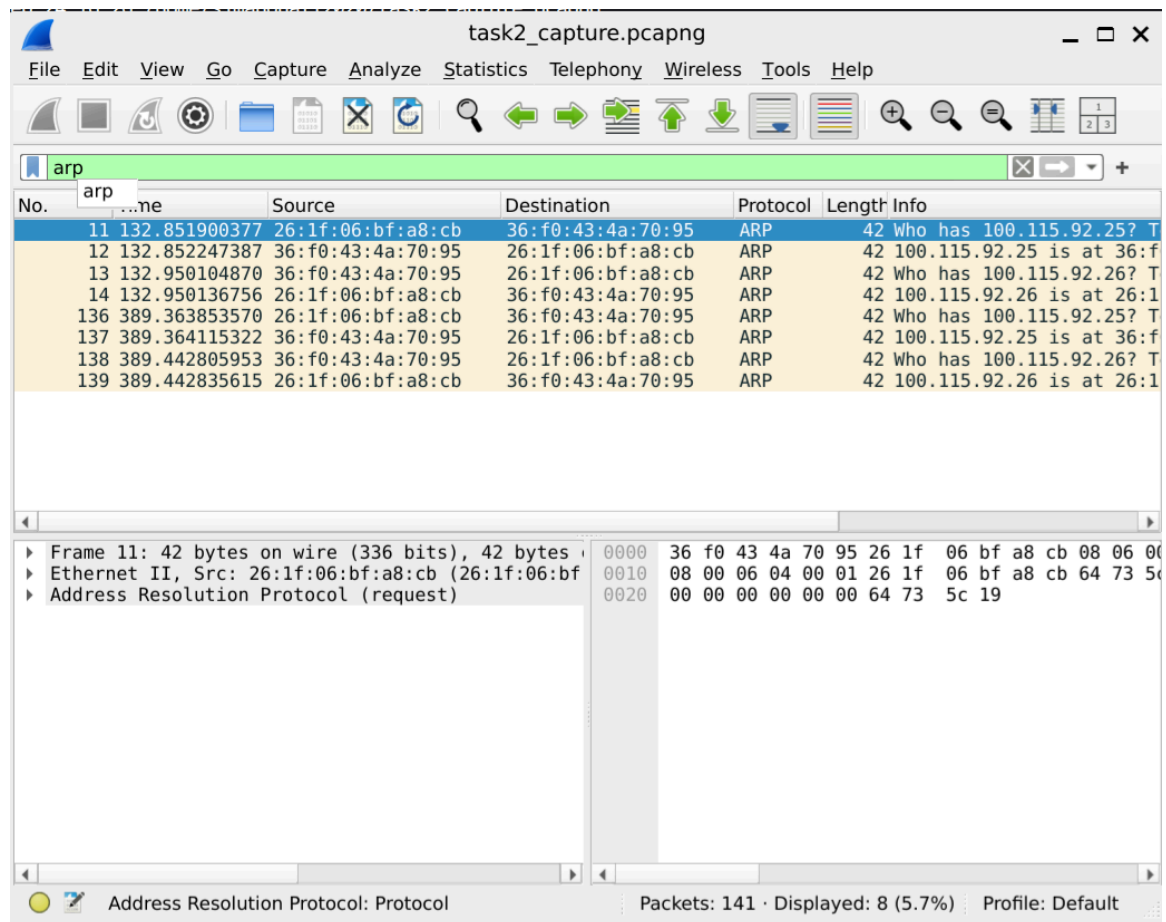
I selected an ARP exchange because it was easy to understand from the captured packets.

Packet 11: The device sends an ARP request asking, "Who has 100.115.92.25?" The source MAC address shown was 26:1f:06:bf:a8:cb.

Packet 12: The device with IP address 100.115.92.25 replies with its MAC address 36:f0:43:4a:70:95.

So, the first packet asks for the MAC address and the second packet gives the answer. This helps the devices communicate on the local network.

Screenshot – ARP packet walkthrough



5. Unencrypted Traffic

I noticed that protocols such as ARP, mDNS and SSDP were visible directly in the capture. Their packet information could be inspected without encryption.

6. What I Learned

I learned how normal network activity creates different types of packets. And also understood how ARP works and how mDNS, NTP and SSDP are used. I also learned how Wireshark filters can be used to find specific protocols.